NOTES

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- ID: 932230195
- ALL PASSWORDS for Administration Building is admin
- ALL PASSWORDS for Engineering Faculty is eng
- ALL PASSWORDS for CS&IT Faculty is cs
- ALL PASSWORDS for Medicine Faculty is med

(THE TELNET PASSWORD is the PASSWORD OF THE VLAN'S SWITCH)

- First, We have the Administration Building from class A which is a one of the 4 buildings we are designing.
- At First we will do the structure of the design then do the VLAN configuration
- For VLAN 10

```
Mostafa-ADMIN(config)#interface range f0/1,f0/2,f0/3,f0/4
Mostafa-ADMIN(config-if-range)#switchport mode access
Mostafa-ADMIN(config-if-range)#switchport access vlan 10
% Access VLAN does not exist. Creating vlan 10
Mostafa-ADMIN(config-if-range)#
```

(VLAN10)

By going to the switch's CLI, We do enable then config terminal. then type interface range and the name of the FastEthernet's port of the PCs connected to the switch. and the switch's port connected to the router.

We do the same process to VLAN 20 and VLAN 30

```
Mostafa-ADMIN>enable
Mostafa-ADMIN#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Mostafa-ADMIN(config)#interface range f0/5,f0/6,f0/7,f0/8
Mostafa-ADMIN(config-if-range)#switchport mode access
Mostafa-ADMIN(config-if-range)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
Mostafa-ADMIN(config-if-range)#
```

(VLAN 20)

```
Mostafa-ADMIN>enable
Mostafa-ADMIN#config termina
Enter configuration commands, one per line.  End with CNTL/Z.
Mostafa-ADMIN(config)#interface range f0/9,f0/10,f0/11,f0/12
Mostafa-ADMIN(config-if-range)#switchport mode access
Mostafa-ADMIN(config-if-range)#switchport access vlan 30
% Access VLAN does not exist. Creating vlan 30
Mostafa-ADMIN(config-if-range)#
```

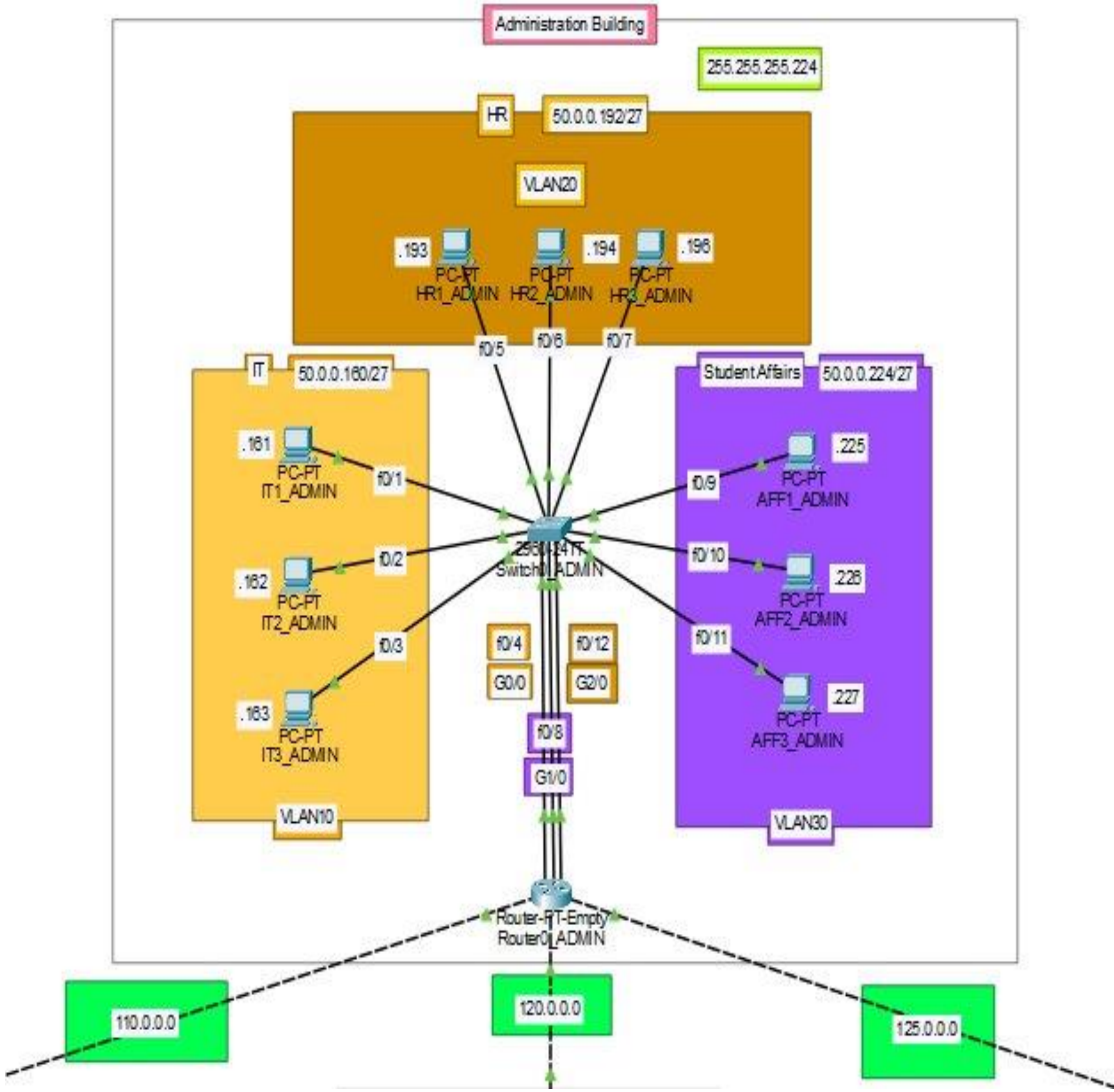
(VLAN 30)

So, VLAN TABLE for Administration Building will be.

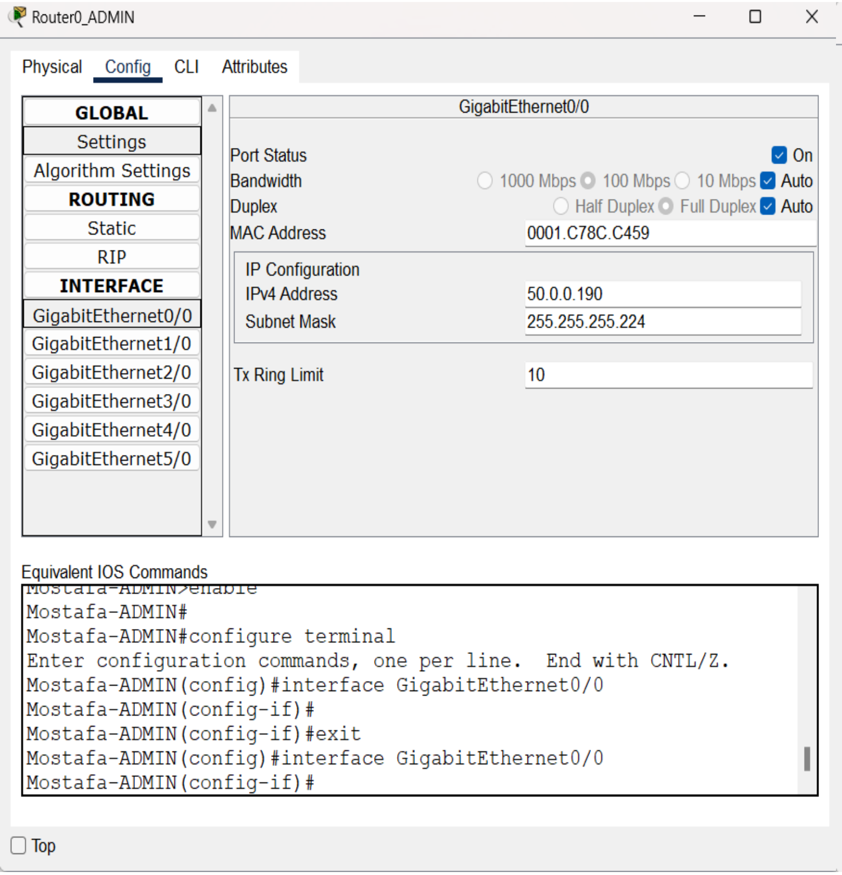
```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   IT_ADMIN                         active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
20   HR_ADMIN                         active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
30   Students_Affairs_ADMIN           active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Mostafa ADMIN#
```

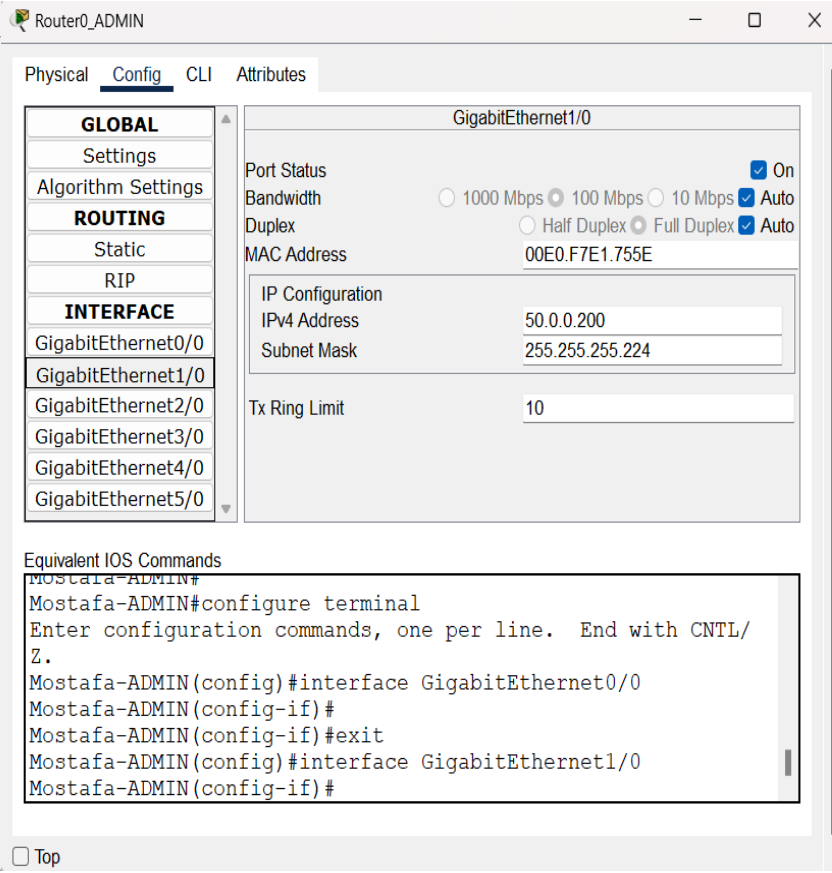
We have now 3 subnetworks. VLAN10, VLAN20 and VLAN30
Connected to a switch then a router. The subnet mask for all networks is 255.255.255.224 And a prefix number /27 which means the host octets has a number of hosts up to 30 hosts for each network we have according to the 2 ^ (number of hosts' bits) – 2
So, 2 ^ (5) – 2 = 30 usable hosts.

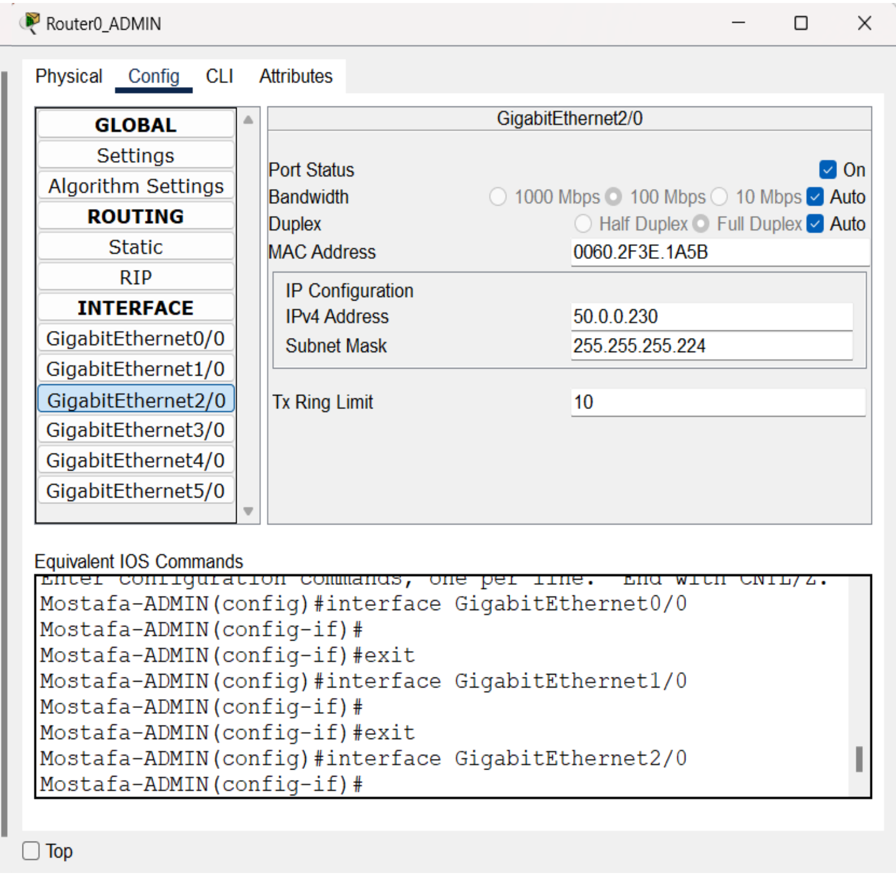
We are starting with IT network which has an IP Address 50.0.0.160
And its PCs' each one has a unique IP in the network Starts from 50.0.0.161 for the first PC to 50.0.0.163 For the third PC.
Then we have the HR network. It has an IP Address 50.0.0.192 It's network IP end with .192 because we've added .160 (which is the last octet for the IT network) + 32 30 hosts for the IT + 1 broadcasting IP and we add extra one for the next network IP Address And we left with .192
As we did with the IT we will do for HR and Affairs networks. Each PC for each network will connect to a central switch using a straight through medium then connected to a router.



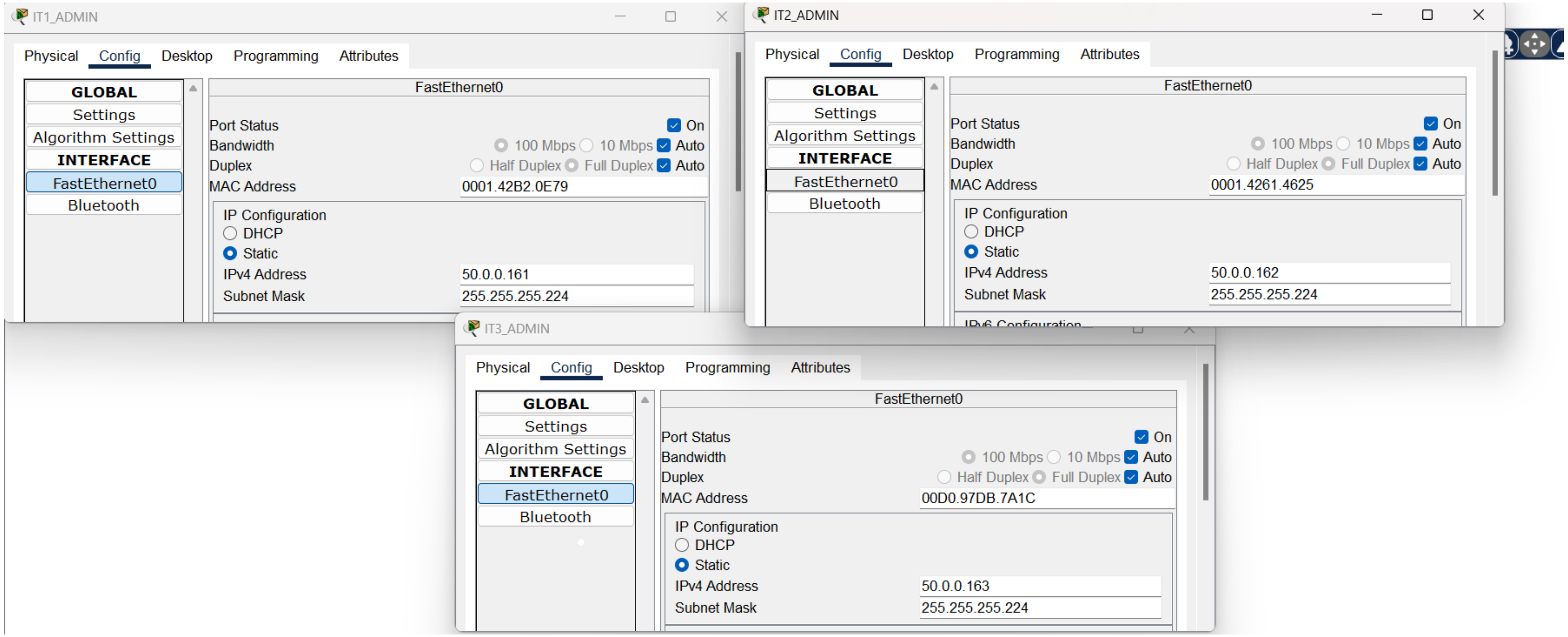
For each port of the router it has its IP Address which we assign to the pc's Gateway.
Each VLAN has its Gateway according to the router's port.
We've here the IP Addresses of all the 3 subnetworks we have
GigabitEthernet0/0 IP Address for PCs in VLAN 10
GigabitEthernet1/0 IP Address for PCs in VLAN 20
GigabitEthernet2/0 IP Address for PCs in VLAN 30



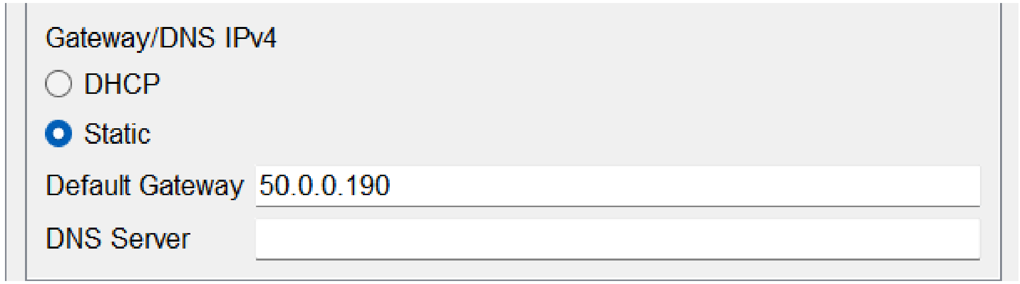




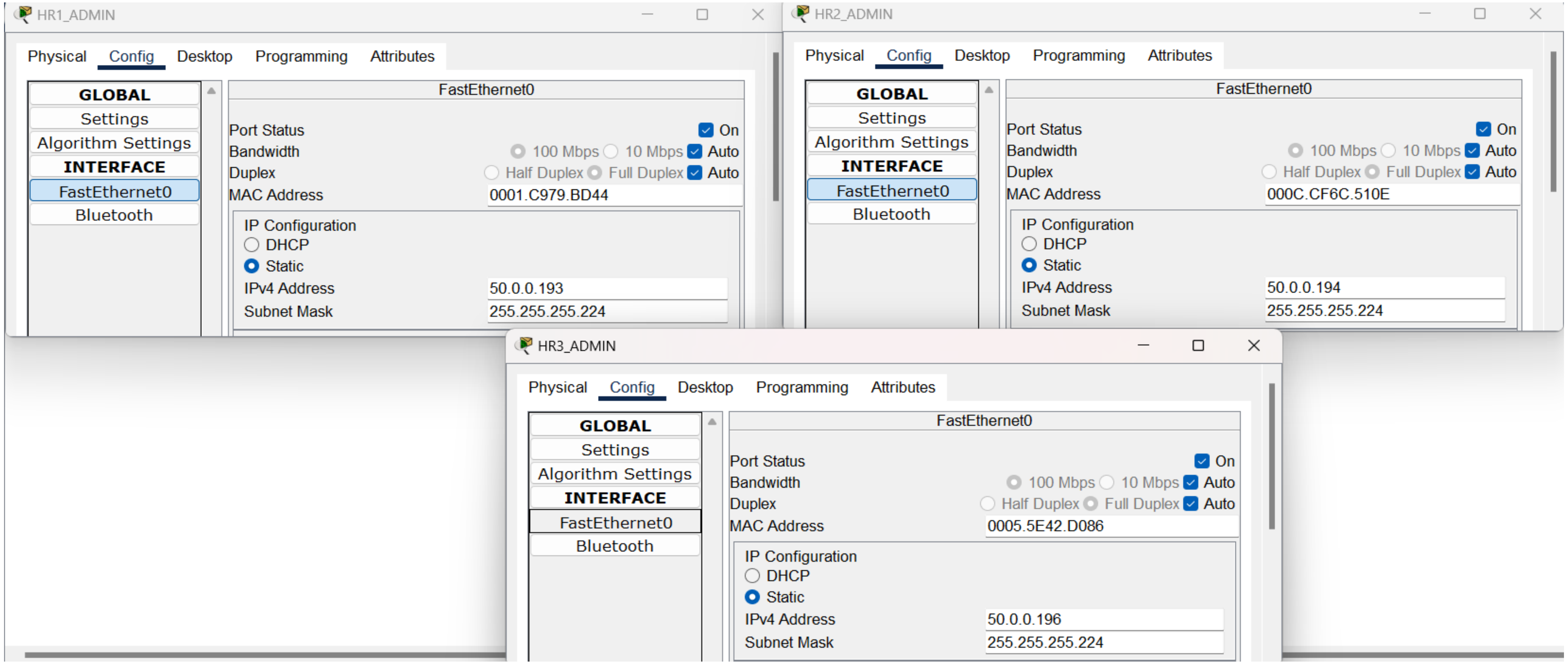
PCs IP Addresses for each PC In VLAN 10



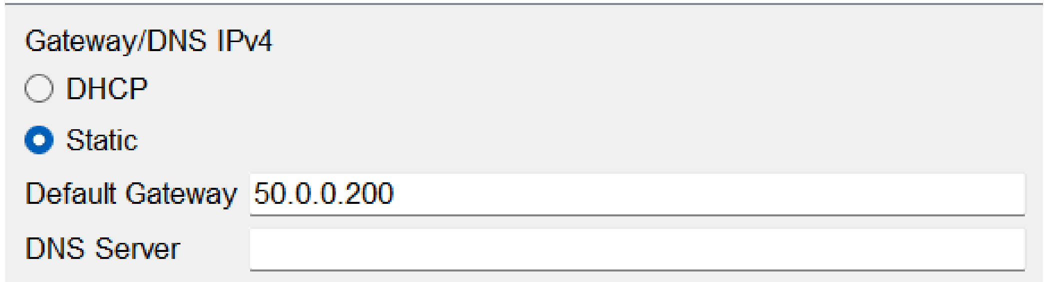
All of them sharing the same Gateway >>>



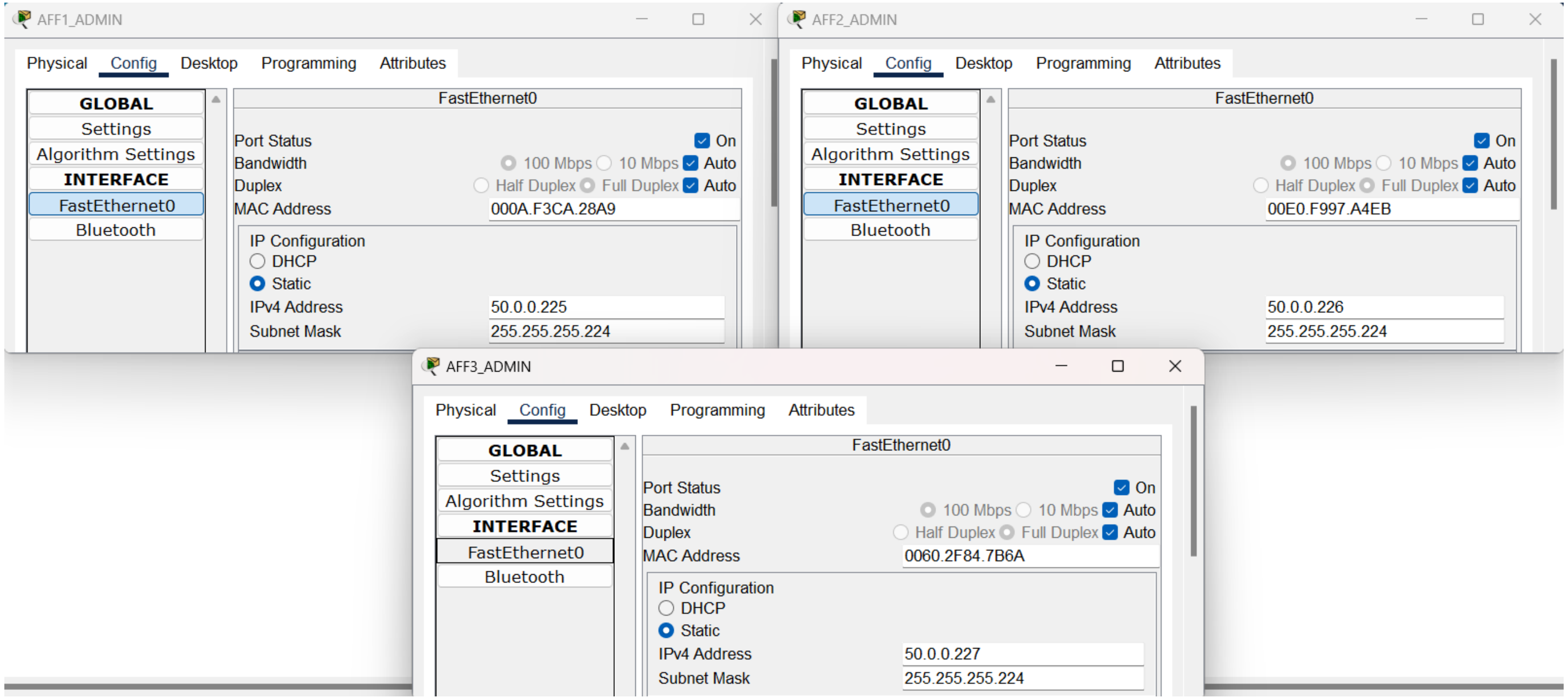
PCs IP Addresses for each PC In VLAN 20



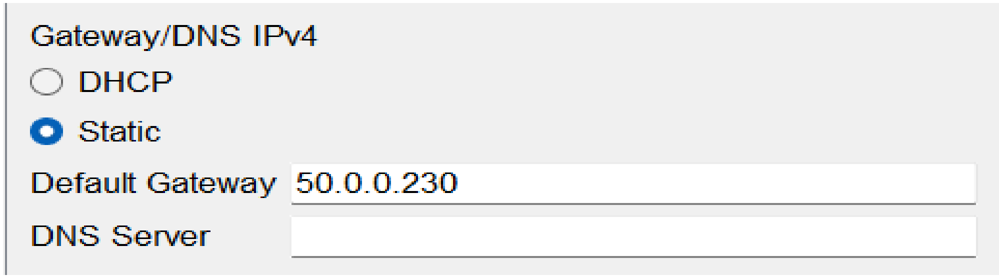
All of them sharing the same Gateway >>>



PCs IP Addresses for each PC In VLAN 30



All of them sharing the same Gateway >>>

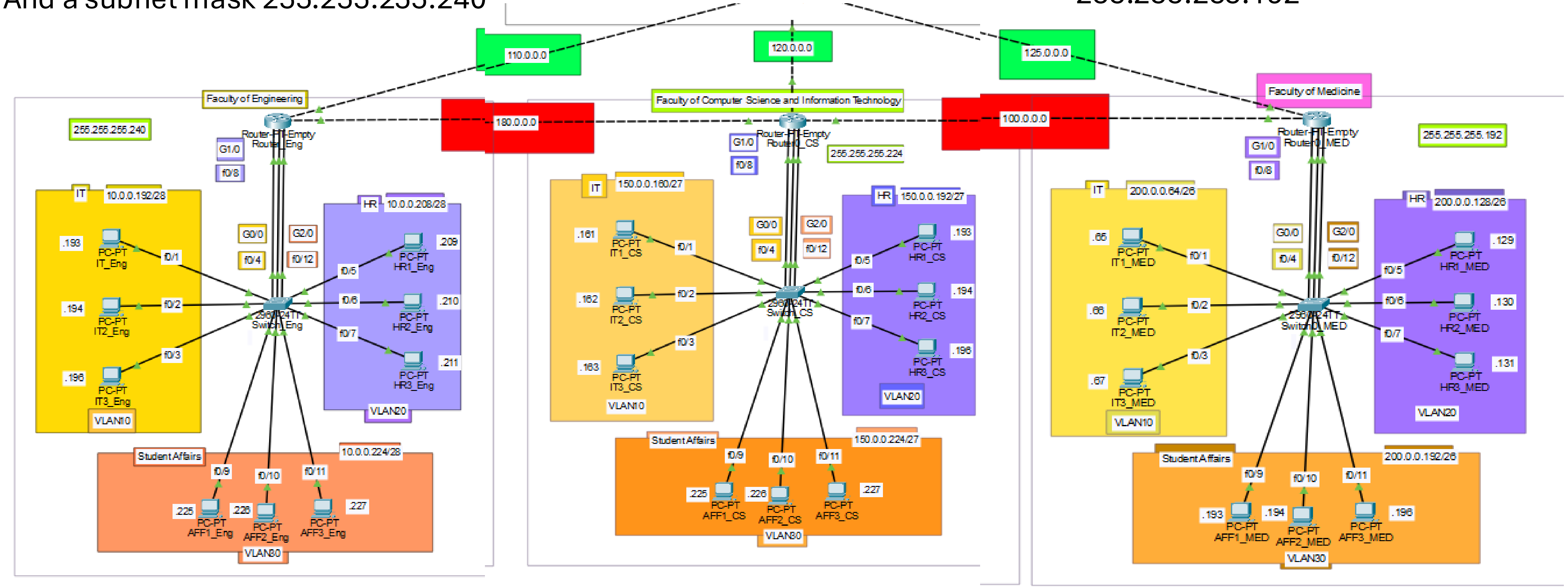


We do the same processes to the other buildings with different things in classes like CS& MEDECINE Buildings and obviously different IPs, subnet masks and prefix number for each Building.

In Engineering faculty we have prefix number /28
And a subnet mask 255.255.255.240

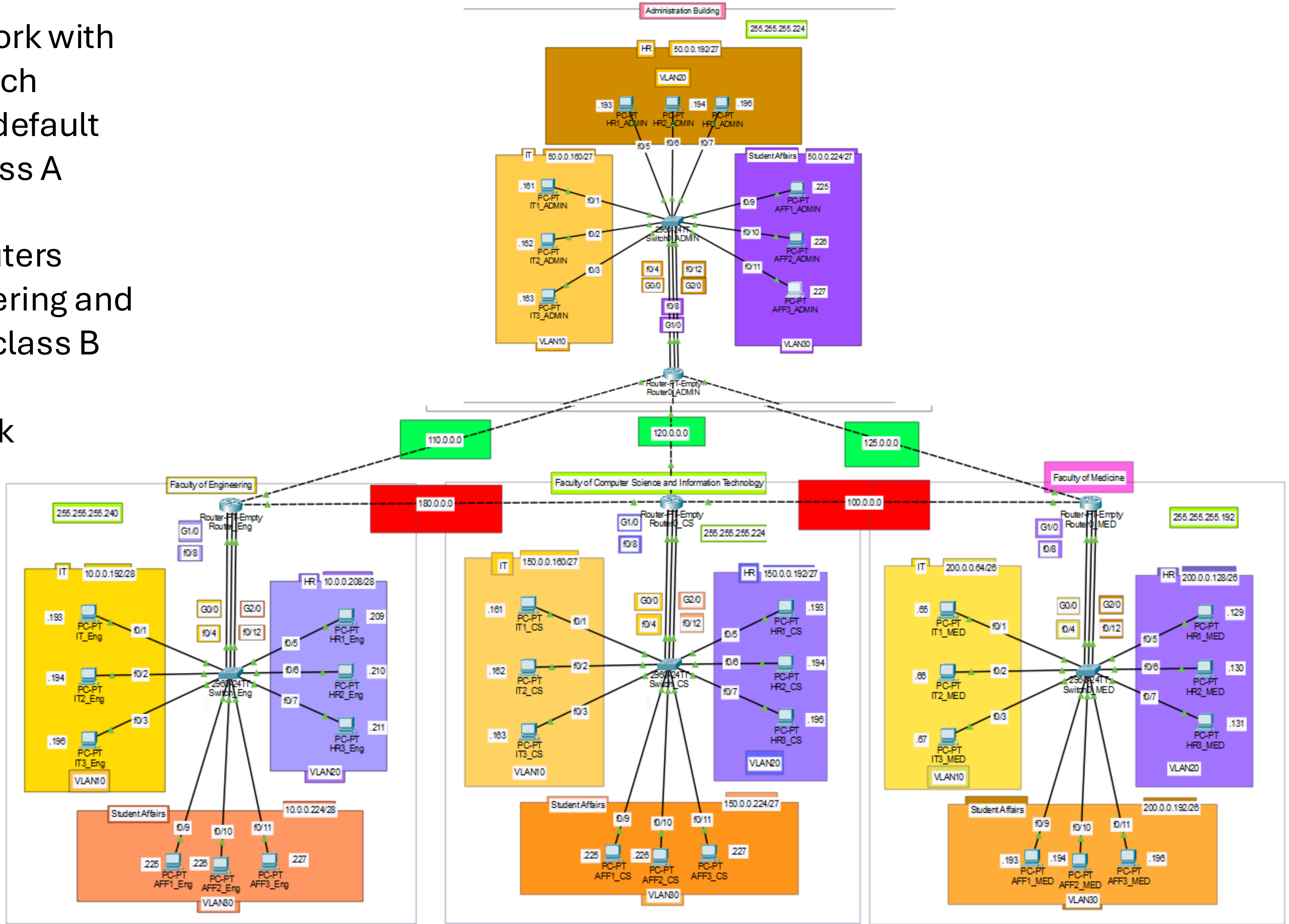
In CS we have class B

In Medicine we have class C, with prefix number /26
And subnet mask 255.255.255.192



NOTE
Each router connected to another router with a cross over medium and establishing a network with an IP Address for each router. And with its default subnet mask for class A which is
255.0.0.0 for all Routers
Except both Engineering and CS&IT Routers are class B
and
Have a subnet mask
255.255.0.0

FINAL TOPOLOGY



VLANs table for each faculty

Engineering Faculty

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   IT_ENG                           active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
20   HR_ENG                           active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
30   Student_Affairs_ENG              active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Mostafa_ENG(config)#
```

CS&IT Faculty

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   IT_CS                            active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
20   HR_CS                            active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
30   Student_Affairs_CS               active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Mostafa_CS#
```

Medicine Faculty

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   IT_MED                           active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
20   HR_MED                           active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
30   Student_Affairs_MED              active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Mostafa_MED(config)#
```

Using static Routing between Engineering and CS&IT faculties Routers

Static Routing:

ip route <network's ip for the next hop router> <subnet mask> <port's ip which will receive the packets >

```
Mostafa-Engineering(config)#ip route 150.0.0.160 255.255.255.224 180.0.0.2
Mostafa-Engineering(config)#ip route 150.0.0.192 255.255.255.224 180.0.0.2
Mostafa-Engineering(config)#ip route 150.0.0.224 255.255.255.224 180.0.0.2
Mostafa-Engineering(config)#
```

```
Mostafa-CS&IT(config)#ip route 10.0.0.192 255.255.255.240 180.0.0.1
Mostafa-CS&IT(config)#ip route 10.0.0.208 255.255.255.240 180.0.0.1
Mostafa-CS&IT(config)#ip route 10.0.0.224 255.255.255.240 180.0.0.1
Mostafa-CS&IT(config)#
```

Using Dynamic Routing (RIP) for all the Routers

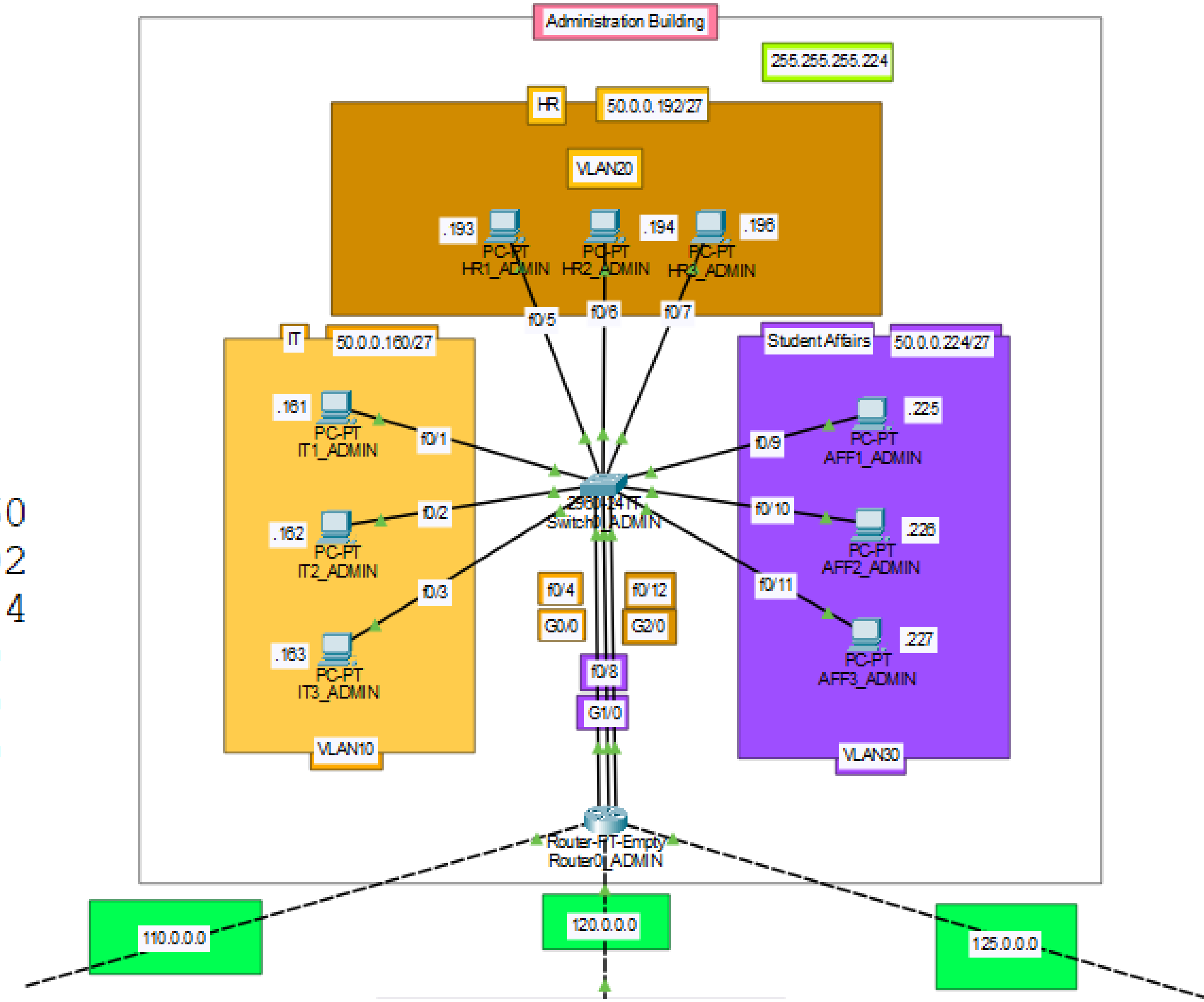
Dynamic Routing (RIP Protocol):

router rip

version 2

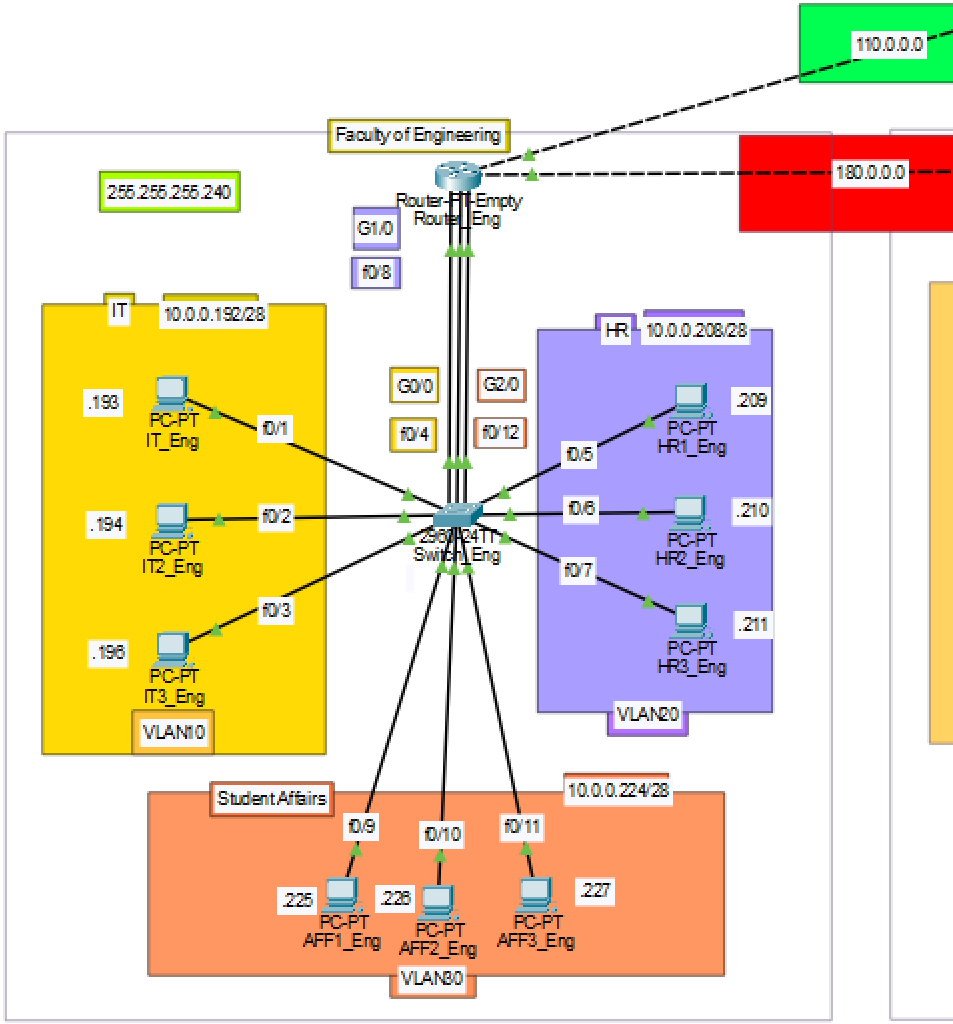
<network> <network's IP directly connected>

```
Mostafa-ADMIN(config)#router rip
Mostafa-ADMIN(config-router)#version 2
Mostafa-ADMIN(config-router)#network 50.0.0.160
Mostafa-ADMIN(config-router)#network 50.0.0.192
Mostafa-ADMIN(config-router)#network 50.0.0.224
Mostafa-ADMIN(config-router)#network 110.0.0.0
Mostafa-ADMIN(config-router)#network 120.0.0.0
Mostafa-ADMIN(config-router)#network 125.0.0.0
Mostafa-ADMIN(config-router)#
```



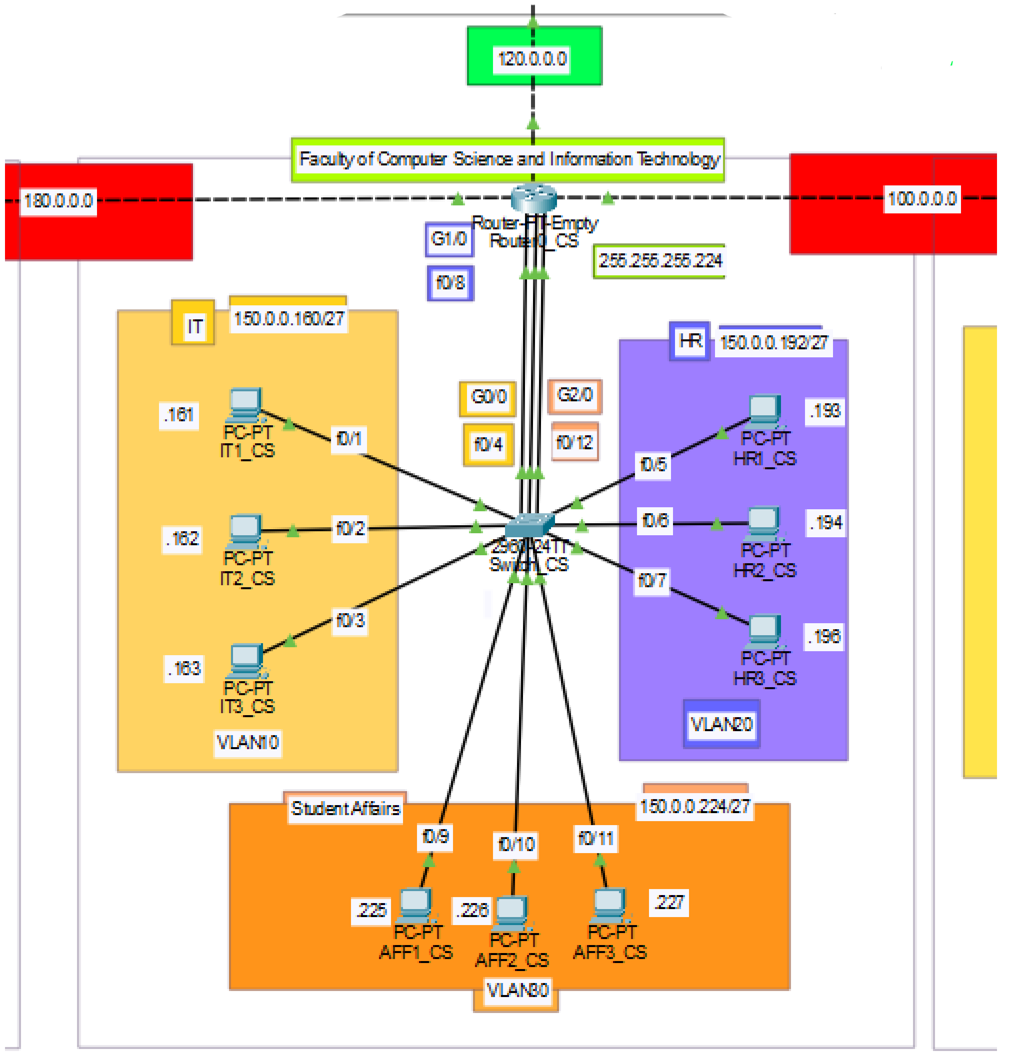
Continue...

```
Mostafa-Engineering(config)#router rip
Mostafa-Engineering(config-router)#version 2
Mostafa-Engineering(config-router)#network 10.0.0.192
Mostafa-Engineering(config-router)#network 10.0.0.208
Mostafa-Engineering(config-router)#network 10.0.0.224
Mostafa-Engineering(config-router)#network 180.0.0.0
Mostafa-Engineering(config-router)#network 110.0.0.0
Mostafa-Engineering(config-router)#
```



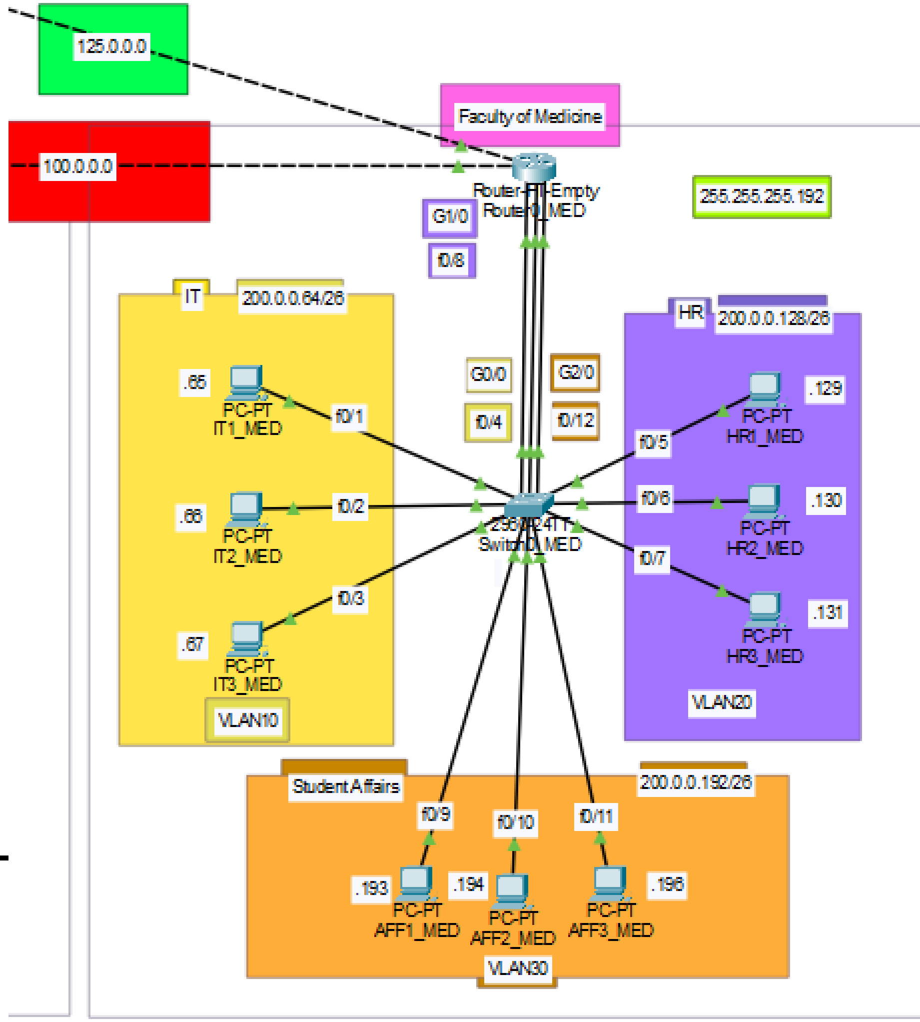
Continue...

```
Mostafa-CS&IT(config)#router rip
Mostafa-CS&IT(config-router)#version 2
Mostafa-CS&IT(config-router)#network 150.0.0.160
Mostafa-CS&IT(config-router)#network 150.0.0.192
Mostafa-CS&IT(config-router)#network 150.0.0.224
Mostafa-CS&IT(config-router)#network 100.0.0.0
Mostafa-CS&IT(config-router)#network 120.0.0.0
Mostafa-CS&IT(config-router)#network 180.0.0.0
Mostafa-CS&IT(config-router)#
```



Continue...

```
Mostafa-Medicine(config)#router rip
Mostafa-Medicine(config-router)#version 2
Mostafa-Medicine(config-router)#network 200.0.0.64
Mostafa-Medicine(config-router)#network 200.0.0.128
Mostafa-Medicine(config-router)#network 200.0.0.192
Mostafa-Medicine(config-router)#network 100.0.0.0
Mostafa-Medicine(config-router)#network 125.0.0.0
Mostafa-Medicine(config-router)#
```



ROUTING TABLE for all Routers

Administration Building

Routing Table for Router0_ADMIN

Type	Network	Port	Next Hop IP	Metric
R	10.0.0.0/8	GigabitEthernet3/0	110.0.0.2	120/1
C	50.0.0.160/27	GigabitEthernet0/0	---	0/0
C	50.0.0.192/27	GigabitEthernet1/0	---	0/0
C	50.0.0.224/27	GigabitEthernet2/0	---	0/0
R	100.0.0.0/8	GigabitEthernet4/0	120.0.0.2	120/1
R	100.0.0.0/8	GigabitEthernet5/0	125.0.0.2	120/1
C	110.0.0.0/8	GigabitEthernet3/0	---	0/0
C	120.0.0.0/8	GigabitEthernet4/0	---	0/0
C	125.0.0.0/8	GigabitEthernet5/0	---	0/0
R	150.0.0.0/16	GigabitEthernet4/0	120.0.0.2	120/1
R	180.0.0.0/16	GigabitEthernet4/0	120.0.0.2	120/1
R	180.0.0.0/16	GigabitEthernet3/0	110.0.0.2	120/1
R	200.0.0.0/24	GigabitEthernet5/0	125.0.0.2	120/1

Engineering Faculty

Routing Table for Router_Eng

Type	Network	Port	Next Hop IP	Metric
C	10.0.0.192/28	GigabitEthernet0/0	---	0/0
C	10.0.0.208/28	GigabitEthernet1/0	---	0/0
C	10.0.0.224/28	GigabitEthernet2/0	---	0/0
R	50.0.0.0/8	GigabitEthernet4/0	110.0.0.1	120/1
R	100.0.0.0/8	GigabitEthernet3/0	180.0.0.2	120/1
C	110.0.0.0/8	GigabitEthernet4/0	---	0/0
R	120.0.0.0/8	GigabitEthernet3/0	180.0.0.2	120/1
R	120.0.0.0/8	GigabitEthernet4/0	110.0.0.1	120/1
R	125.0.0.0/8	GigabitEthernet4/0	110.0.0.1	120/1
R	150.0.0.0/16	GigabitEthernet3/0	180.0.0.2	120/1
S	150.0.0.160/27	---	180.0.0.2	1/0
S	150.0.0.192/27	---	180.0.0.2	1/0
S	150.0.0.224/27	---	180.0.0.2	1/0
C	180.0.0.0/16	GigabitEthernet3/0	---	0/0
R	200.0.0.0/24	GigabitEthernet3/0	180.0.0.2	120/2
R	200.0.0.0/24	GigabitEthernet4/0	110.0.0.1	120/2

CS&IT Faculty

Routing Table for Router0_CS

Type	Network	Port	Next Hop IP	Metric
R	10.0.0.0/8	GigabitEthernet3/0	180.0.0.1	120/1
S	10.0.0.192/28	---	180.0.0.1	1/0
S	10.0.0.208/28	---	180.0.0.1	1/0
S	10.0.0.224/28	---	180.0.0.1	1/0
R	50.0.0.0/8	GigabitEthernet5/0	120.0.0.1	120/1
C	100.0.0.0/8	GigabitEthernet4/0	---	0/0
R	110.0.0.0/8	GigabitEthernet5/0	120.0.0.1	120/1
R	110.0.0.0/8	GigabitEthernet3/0	180.0.0.1	120/1
C	120.0.0.0/8	GigabitEthernet5/0	---	0/0
R	125.0.0.0/8	GigabitEthernet4/0	100.0.0.2	120/1
R	125.0.0.0/8	GigabitEthernet5/0	120.0.0.1	120/1
C	150.0.0.160/27	GigabitEthernet0/0	---	0/0
C	150.0.0.192/27	GigabitEthernet1/0	---	0/0
C	150.0.0.224/27	GigabitEthernet2/0	---	0/0
C	180.0.0.0/16	GigabitEthernet3/0	---	0/0
R	200.0.0.0/24	GigabitEthernet4/0	100.0.0.2	120/1

Medicine Faculty

Routing Table for Router0_MED

Type	Network	Port	Next Hop IP	Metric
R	10.0.0.0/8	GigabitEthernet3/0	100.0.0.1	120/2
R	10.0.0.0/8	GigabitEthernet4/0	125.0.0.1	120/2
R	50.0.0.0/8	GigabitEthernet4/0	125.0.0.1	120/1
C	100.0.0.0/8	GigabitEthernet3/0	---	0/0
R	110.0.0.0/8	GigabitEthernet4/0	125.0.0.1	120/1
R	120.0.0.0/8	GigabitEthernet3/0	100.0.0.1	120/1
R	120.0.0.0/8	GigabitEthernet4/0	125.0.0.1	120/1
C	125.0.0.0/8	GigabitEthernet4/0	---	0/0
R	150.0.0.0/16	GigabitEthernet3/0	100.0.0.1	120/1
R	180.0.0.0/16	GigabitEthernet3/0	100.0.0.1	120/1
C	200.0.0.64/26	GigabitEthernet0/0	---	0/0
C	200.0.0.128/26	GigabitEthernet1/0	---	0/0
C	200.0.0.192/26	GigabitEthernet2/0	---	0/0

S for Static Routing   *R for Dynamic Routing (RIP)*   *C for directly connected networks*

TELNET CONFIGURATION

Administration Building

```
Mostafa_ADMIN(config)#
Mostafa_ADMIN(config)#interface vlan 20
Mostafa_ADMIN(config-if)#ip address 50.0.0.195 255.255.255.224
Mostafa_ADMIN(config-if)#no shutdown
Mostafa_ADMIN(config-if)#
```

Engineering Faculty

```
Mostafa_ENG(config)#interface vlan 10
Mostafa_ENG(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

Mostafa_ENG(config-if)#ip address 10.0.0.195 255.255.255.240
Mostafa_ENG(config-if)#no shutdown
Mostafa_ENG(config-if)#
```

CS&IT Faculty

```
Mostafa_CS(config)#
Mostafa_CS(config)#interface vlan 20
Mostafa_CS(config-if)#ip address 150.0.0.195 255.255.255.224
Mostafa_CS(config-if)#no shutdown
Mostafa_CS(config-if)#
```

Medicine Faculty

```
Mostafa_MED(config)#interface vlan 30
Mostafa_MED(config-if)#ip address 200.0.0.195 255.255.255.192
Mostafa_MED(config-if)#no shutdown
```

THE END

Hope you like it.

By: Mostafa Ashraf Suliman

Bonus?